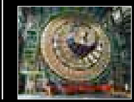
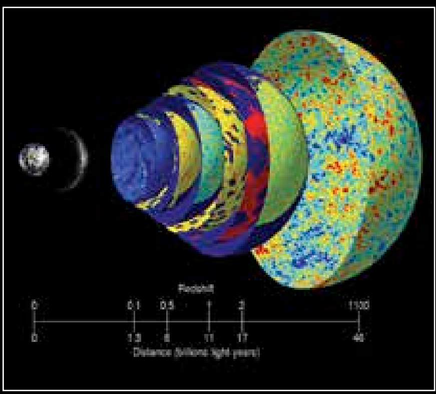


The hunt continues
The Large Hadron Collider resumes its search for more unknown particles. Read on **http://dgit.in/lhcworks**

mathematics behind this explanation has already been vigorously tested and is part of a field called holography.

Unlike the popularised understanding of holography such as in holographic displays, the study of the holographic principle is related to string theory and quantum mechanics. It studies the properties of strings and is also related to the study of quantum gravity.

Fortunately, like any good scientific theory, the theory of multi-dimensional brane universes is easy to test. Since the fourth dimensional matter would undergo thermal fluctuations due to the four dimensional black hole it would register as small but noticeable distortions in the cosmic microwave radiation - a change we can observe and measure. If these distortions aren't in line with the new theory, then it would mean the new theory is wrong. Other



Cosmic microwave background radiation lets us observe the very origins of the universe.

similar aspects of the new theory can be tested and measured proving its veracity. All it takes is a little time and a really big cosmic radiation detector. We'll know soon enough if this theory is right or not.

Ad infinitum

While the theory of a five dimensional universe takes a while to digest - what about the theory that the universe is infinite in age with no beginning or end? In another competing new theory that is coming to the attention of scientists the currently accepted age of the universe - 13.8 billion years - may be off by, well, an infinity or so. According to this new model which uses quantum corrections within Einstein's theory of general relativity, we find an elegant answer to mysteries such as dark matter, dark energy and so much more.

The new model was postulated by Egyptian scientist Ahmed Farag Ali and Indian scientist Saurya Das. These two researches used the foundation work of 1950s theoretical physicist David Bohm to replace the use of classical geodesics (shortest path between two points on a curved surface) with the newly developed method of quantum trajectories between two points. They applied the quantum Bohmian trajectories to an equation by physicist Amal Kumar Raychaudhuri of Presidency College in Calcutta to arrive at their final model explaining the design of the universe.

Within their mashup work, by combining the ideas and approaches of geniuses across the world and time, the duo of Ali and Das was able to derive an equation which described the expansion and evolution of the known universe. Their equation combines the properties of quantum theory with general relativity which are expected to hold true, if and when a unified theory of quantum gravity is discovered.

Using the new variables and factors in their equation, as a representation of the universe, these scientists have posited that the universe is of a finite size. This conclusion also has the natural corollary that the universe is of an infinite age implying that time has neither a beginning nor an end. The reason this theory really holds up is because it is consistent and closely congruent with the most recent observations regarding the density of the universe and the cosmological constant. So far it seems to check all the boxes making it a likely contender for the universe's origin story - which is that it has none.

Unquenchable thirst

The nature of reality is perhaps impossible to discover. What came before the big bang? Perhaps we will never know. Was there a big bang? We might be able to answer that eventually. What does it all mean? We don't know, and perhaps never will. That's not going to stop us from continuing to ask questions, and when every answer throws up a whole new set of questions, you can bet humanity is going to thirst for answers for a long, long time to come.